GAME EDUKASI · UNITY 3D

EduQuest

Jelajah Sistem Organ Manusia

Tutorial Membangun Tampilan di Unity — Lengkap dengan
Posisi, Skala, Warna, Kamera, Lampu & Layout UI

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Mata Kuliah: Praktik Pengembangan Game Pembelajaran

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Daftar Isi

> Semua angka di tutorial ini adalah nilai **Transform** (Position / Rotation / Scale), **warna** (masukkan kode hex di kolom Hex pada color picker Unity), dan **ukuran UI** (RectTransform). Fokus dokumen ini hanya **tampilan** — script/kode dibahas terpisah.

1. Persiapan & Catatan Penggunaan

- Project: Unity 6, template **3D (URP)**.
- Package: **Input System**, **TextMeshPro** (Import TMP Essentials), **Cinemachine**, **Starter Assets - ThirdPerson**.
- Satuan posisi = meter. Untuk memasukkan warna: klik kotak warna > pada field **Hexadecimal** ketik kodenya.
- Membuat objek 3D: GameObject > 3D Object > (Cube/Sphere/Cylinder). Membuat UI: GameObject > UI > ...
- Anchor UI: di RectTransform klik kotak **Anchor Presets**. "TL"=kiri-atas, "TR"=kanan-atas, "TC"=tengah-atas, "BC"=tengah-bawah, "C"=tengah.

2. Palet Warna (Referensi Hex)

Tiap level punya warna dasar; warna lain adalah turunannya (lebih gelap/terang). Pakai tabel ini saat membuat Material atau mengatur warna lampu/fog.

Pemakaian	Jantung	Paru-paru	Otak
Warna dasar organ	#8B0000	#4FC3F7	#7B1FA2
Material dinding (×0.85)	#760000	#43A6D2	#681B8A
Aksen / ceiling / pilar (×0.55)	#4C0000	#2B6B88	#441159
Lantai (lebih gelap)	#3E0000	#23586F	#370E49
Warna Fog	#5A0000	#3380A1	#501569
Bola stasiun / glow (lebih terang)	#B35A5A	#8DD8FA	#A96DC3
Organ aksen (m2)	#A84040	#7BD2F9	#9C57B9

UI / lainnya	Hex / RGBA
Background Menu (kamera)	#0D0D14
Panel Menu	#0F0F1A
Overlay panel quiz/pause	Hitam, Alpha 0.85 (RGBA 0,0,0,217)
Kotak quiz / feedback	#1F1F2E
Kotak result	#1A2433
Tombol jawaban	#E6E6F2 (teks hitam)
Warna nyawa (Lives)	#FF4D4D
Teks skor result	#FFE666
Collectible (item)	#FFD933

3. Scene Main Menu — Layout UI

New Scene > simpan **MainMenu**. Tambah **Main Camera** (Clear Flags = Solid Color, Background **#0D0D14**). Buat **Canvas** (Render Mode Screen Space - Overlay; Canvas Scaler = Scale With Screen Size; Reference Resolution **1920×1080**; Match **0.5**).

3.1 MainPanel (Image, warna #0F0F1A, stretch penuh)

Elemen	Anchor	Pos (X,Y)	Width×Height	Font	Isi / Warna
Judul (TMP)	TC	0, −120	1200×130	90	"EduQuest" putih
Subjudul (TMP)	TC	0, −240	1200×60	40	"Jelajah Sistem Organ Manusia" #B3D9FF
Button Main	C	0, 60	450×80	—	teks "Main"
Button Pengaturan	C	0, −40	450×80	—	teks "Pengaturan"
Button Tentang	C	0, −140	450×80	—	teks "Tentang"
Button Keluar	C	0, −240	450×80	—	teks "Keluar"

3.2 SettingsPanel (Image #0F0F1A, nonaktif di awal)

Elemen	Anchor	Pos	Size	Font
Judul "Pengaturan"	TC	0, −140	1000×100	70
Label "Volume"	C	0, 80	600×60	40
Slider Volume	C	0, 10	600×30	—
Info kontrol (teks)	C	0, −120	900×120	28
Button Kembali	C	0, −320	450×80	—

3.3 AboutPanel (Image #0F0F1A, nonaktif di awal)

Elemen	Anchor	Pos	Size	Font
Judul "Tentang Game"	TC	0, −120	1000×100	70
Deskripsi (teks)	C	0, 20	1100×500	30
Button Kembali	C	0, −360	450×80	—

Objek non-visual: Empty **GameManagers** & **MainMenuController** (script dibahas terpisah). Tombol jawaban/menu = Button - TextMeshPro (punya komponen Image).

4. Scene Level — Lingkungan, Fog & Skybox

New Scene > simpan **Level1_Jantung**. Atur pencahayaan & suasana (Window > Rendering > Lighting > Environment).

4.1 Directional Light

Properti	Nilai
Rotation	X 50, Y −30, Z 0
Intensity	0.5
Color	Jantung #DCB3B3 · Paru #C7ECFC · Otak #C9A9D9 (warna organ dicampur putih)
Shadow Type	Soft Shadows

4.2 Fog (Lighting > Environment > Other Settings)

Properti	Nilai
Fog	Aktif (centang)
Mode	Exponential Squared
Color	Jantung #5A0000 · Paru #3380A1 · Otak #501569
Density	Jantung 0.045 · Paru 0.028 · Otak 0.035

4.3 Ambient & Skybox

Properti	Nilai
Environment Lighting Source	Color (Flat)
Ambient Color	Jantung #530000 · Paru #2E6D89 · Otak #3D0F50
Skybox Material	Buat Material shader **Skybox/Procedural**: Sky Tint = warna organ, Ground = warna lantai, Exposure 0.7, Atmosphere Thickness 0.5

5. Scene Level — Ruangan

Buat Empty **Room_Jantung** (jadikan parent semua bagian). Ruangan ukuran 20 (lebar X) × 36 (panjang Z) × 6 (tinggi Y). Buat Material: dinding, aksen, lantai (lihat palet warna bab 2).

5.1 Lantai, Ceiling, Dinding (Cube)

Objek	Position (X,Y,Z)	Scale (X,Y,Z)	Material
Floor	0, −0.2, 0	20, 0.4, 36	Lantai (#3E0000)
Ceiling	0, 6.2, 0	20, 0.4, 36	Aksen (#4C0000)
Wall_North	0, 3, 18	20, 6, 0.4	Dinding (#760000)
Wall_South	0, 3, −18	20, 6, 0.4	Dinding
Wall_East	10, 3, 0	0.4, 6, 36	Dinding
Wall_West	−10, 3, 0	0.4, 6, 36	Dinding

5.2 Lampu Ruangan (4× Point Light)

Range 16, Intensity 2.2, Color warna organ dicampur putih (mis. Jantung #CB8C8C).

Lampu	Position
RoomLight_0	0, 5.2, −10.8
RoomLight_1	0, 5.2, −3.6
RoomLight_2	0, 5.2, 3.6
RoomLight_3	0, 5.2, 10.8

5.3 Pilar (8× Cube, scale 0.8 × 6 × 0.8, Material Aksen)

Z	Kiri (X=−8.8)	Kanan (X=8.8)
−13	−8.8, 3, −13	8.8, 3, −13
−5	−8.8, 3, −5	8.8, 3, −5
3	−8.8, 3, 3	8.8, 3, 3
11	−8.8, 3, 11	8.8, 3, 11

5.4 Titik Penting

Objek	Position	Scale
PlayerSpawnPoint (Empty)	0, 0.1, −15.5	—
OrganPlatform (Cube, Aksen)	0, 0.25, 14	6, 0.5, 6
OrganModel_Holder (Empty)	0, 1.8, 14	—

6. Scene Level — Model Organ (Placeholder)

Di dalam **OrganModel_Holder**, susun primitive berikut (posisi/skala bersifat **lokal** terhadap holder). Material utama = warna organ (#8B0000), material aksen = m2 (#A84040). Tambah komponen **Rotator** pada parent organ (Rotation Speed Y = 25).

Jantung

Bentuk	Local Position	Local Scale	Material
Sphere	−0.45, 0.4, 0	1, 1, 1	Organ
Sphere	0.45, 0.4, 0	1, 1, 1	Organ
Sphere (badan)	0, −0.3, 0	1.3, 1.6, 1.3	Organ
Cylinder (arteri)	−0.3, 1.1, 0	0.25, 0.5, 0.25	Aksen
Cylinder (arteri)	0.3, 1.1, 0	0.25, 0.5, 0.25	Aksen

Paru-paru

Bentuk	Local Position	Local Scale
Sphere (lobus)	−0.7, 0, 0	1.1, 1.9, 1.0
Sphere (lobus)	0.7, 0, 0	1.1, 1.9, 1.0
Cylinder (trakea)	0, 1.1, 0	0.3, 0.6, 0.3
Cylinder (bronkus)	−0.35, 0.5, 0	0.18, 0.4, 0.18
Cylinder (bronkus)	0.35, 0.5, 0	0.18, 0.4, 0.18

Otak

Bentuk	Local Position	Local Scale
Sphere (utama)	0, 0.2, 0	1.8, 1.4, 1.6
Sphere (gyrus)	−0.6, 0.7, 0.4	0.6
Sphere (gyrus)	0.6, 0.7, 0.4	0.6
Sphere (gyrus)	0, 0.8, −0.6	0.6
Sphere (cerebellum)	0, −0.5, −0.7	0.7

Label nama organ: 3D Text/TextMesh di local (0, 1.8, 0), Character Size kecil (Scale 0.15), teks "Jantung".

7. Scene Level — Player & Kamera

7.1 Player

Langkah	Detail
Drag prefab	**PlayerArmature** (Starter Assets) ke scene
Position	sama dengan PlayerSpawnPoint: 0, 0.1, −15.5
Rotation	0, 0, 0 (menghadap +Z, ke arah organ)
Tag	**Player**
Komponen	Add **PlayerInteraction** (assign ThirdPersonController, StarterAssetsInputs, PlayerInput)

7.2 Kamera (Cinemachine third-person)

Objek	Pengaturan
Main Camera	Add Component **CinemachineBrain**
PlayerFollowCamera	Drag prefab **PlayerFollowCamera** ke scene
Tracking / Follow Target	set ke **PlayerCameraRoot** (child di dalam PlayerArmature)

Nilai default PlayerFollowCamera Starter Assets sudah pas (jarak/tinggi orbit). Jika ingin atur sendiri: biasanya kamera mengikuti dari belakang-atas, kira-kira offset (0, 3, −6) dari pemain.

8. Scene Level — Stasiun Quiz, Collectible, AI Guide

8.1 Tiga Stasiun Quiz

Buat Empty **QuizStation_1/2/3** pada posisi (world) berikut:

Stasiun	Position	Question Index
QuizStation_1	−5, 0, −6	0
QuizStation_2	5, 0, 1	1
QuizStation_3	−5, 0, 8	2

Isi tiap stasiun (transform **lokal** terhadap stasiun):

Bagian	Bentuk	Local Pos	Local Scale / Setting
Pedestal	Cube	0, 0.5, 0	1.6, 1, 1.6 — Material aksen gelap
Orb	Sphere	0, 2, 0	0.9 — Material glow (#B35A5A); + FloatBob + Rotator(Y40)
StationLight	Point Light	0, 2, 0	Range 7, Intensity 2.5, warna glow
Label	3D Text	0, 3.1, 0	Scale 0.1, teks "Soal 1" + Billboard
Trigger	Box Collider	Center 0,1.5,0	Size 4, 3, 4 — centang **Is Trigger**

Lalu Add Component **QuizStation** (Quiz Data, Question Index, Station Number, Total Stations=3, Glow Renderer=Orb, Active Indicator=StationLight).

8.2 Collectibles (6× Sphere, Scale 0.5, warna #FFD933)

Tambah Sphere Collider (Is Trigger) + komponen Collectible + Rotator (Y 90).

#	Position	#	Position
1	2, 1, −9	4	−2, 1, 11
2	−3, 1, −2	5	6, 1, −4
3	4, 1, 5	6	−6, 1, 3

8.3 AI Guide (dekat spawn)

Bagian	Detail
Posisi (world)	2.5, 1.7, −13.5 — Add FloatBob pada parent
Body (Sphere)	Scale 0.6, Material warna organ+putih; + Rotator (Y30)
Light	Point, Range 5, Intensity 1.6
Label (3D Text)	Local 0,1,0; Scale 0.08; teks "Halo! Temukan 3 stasiun..." + Billboard

9. Scene Level — Canvas UI

Buat **GameCanvas** (Scaler 1920×1080, Match 0.5). Add komponen UIManager & PauseMenu (kosongkan dulu, isi fieldnya nanti).

9.1 HUD

Elemen	Anchor / Pivot	Pos	Size	Font / Align	Warna / Isi
ScoreText	TL	40, −30	400×60	36 / Left	putih "Skor: 0"
LivesText	TR	−40, −30	400×60	40 / Right	#FF4D4D "♥ ♥ ♥"
LevelNameText	TC	0, −30	700×60	42 / Center	putih "Level 1 - Jantung"
StationProgressText	TC	0, −95	500×50	30 / Center	#CCE6FF "Stasiun: 0 / 3"
InteractPrompt	BC	0, 140	800×60	34 / Center	kuning "Tekan [E] untuk berinteraksi"

9.2 Quiz Panel (overlay hitam α0.85)

Di dalamnya kotak "Box" (Image #1F1F2E) di tengah, Size 1100×720. Elemen child anchor ke **TC** kotak:

Elemen	Pos	Size	Font
Progress "Soal 1/3"	0, −30	900×50	30
QuestionText	0, −180	980×220	40
Answer 0	0, −360	880×80	30
Answer 1	0, −455	880×80	30
Answer 2	0, −550	880×80	30
Answer 3	0, −645	880×80	30

Tombol jawaban: Image #E6E6F2, teks hitam.

9.3 Feedback Panel (overlay; kotak #1F1F2E, Size 1000×520)

Elemen	Pos	Size	Font
Title (BENAR/SALAH)	0, −60 (TC)	900×90	60
Explanation	0, 0 (C)	900×240	32
Tombol Lanjut	0, 50 (BC)	400×80	—

9.4 Result Panel (overlay α0.9; kotak #1A2433, Size 1000×760)

Elemen	Pos (TC)	Size	Font / Warna
Title "Hasil Level"	0, −50	900×90	60 putih
Skor Total	0, −170	900×70	44 / #FFE666
Benar	0, −250	900×60	36 / #66FF66
Salah	0, −310	900×60	36 / #FF8080
Motivasi	0, −400	920×120	30 / #D9F2FF
Tombol Level Berikutnya	0, 230 (BC)	500×75	—
Tombol Ulangi	0, 145 (BC)	500×75	—
Tombol Menu Utama	0, 60 (BC)	500×75	—

9.5 Pause Panel (Canvas terpisah, Sort Order 20; overlay α0.85)

Elemen	Anchor	Pos	Size	Font
Title "JEDA"	TC	0, −180	800×120	80
Button Lanjutkan	C	0, 60	450×80	—
Button Ulangi Level	C	0, −40	450×80	—
Button Menu Utama	C	0, −140	450×80	—

> Sembunyikan (uncheck) Quiz/Feedback/Result/Pause panel & InteractPrompt di awal. Fade: buat Canvas hitam full-screen (Sort Order 100) + CanvasGroup (Alpha 0).

10. Scene EndScreen

New Scene > **EndScreen**. Main Camera Background **#0A0F0A**. Canvas + Panel (Image #0F170F).

Elemen	Anchor	Pos	Size	Font / Warna
Title "SELAMAT!"	TC	0, −120	1200×130	90 putih
Skor Akhir	C	0, 160	900×80	55 / #FFE666
Statistik	C	0, 80	900×60	40 putih
Pesan	C	0, −40	1000×160	32 / #D9F2D9
Button Main Lagi	C	0, −260	450×80	—
Button Menu Utama	C	0, −360	450×80	—

Objek non-visual: GameManagers, EndScreenController.

11. Build Settings

File > Build Settings > Add Open Scenes berurutan:

> **MainMenu → Level1_Jantung → Level2_ParuParu → Level3_Otak → EndScreen**

Untuk WebGL: Switch Platform ke WebGL, Color Space **Gamma**, Compression **Gzip**.

— Selesai. Tutorial tampilan lengkap. —